Toulouse School of Economics, 2012-2013
Macroeconomics 1 – Franck Portier

Problem set 4: speed of convergence

Problem I – Log-linearizing the Ramsey model around the steady state and computing its speed of convergence

Let us consider the following dynamic equations of a Ramsey model without long-term growth:

$$\begin{aligned}\frac{\dot{C}_t}{C_t} &= \eta\left[f'(K_t) - \delta - \rho\right];\\ \dot{K}_t &= f(K_t) - \delta K_t - C_t.\end{aligned}$$

We assume that η is a constant and that $f(K_t) \equiv AK_t^\alpha$.

1 – What are the steady state values of K and C, denoted by $K^\star$ and $C^\star$?

We now assume we are close to the steady state, with $C_t = C^\star(1+\hat{c}_t)$, $K_t = K^\star(1+\hat{k}_t)$, $\hat{c}_t, \hat{k}_t, \frac{d\hat{c}_t}{dt}, \frac{d\hat{k}_t}{dt} << 1$.

2 – Show that there is a matrix M such that

$$\begin{pmatrix}\frac{d\hat{k}_t}{dt}\\ \frac{d\hat{c}_t}{dt}\end{pmatrix} \approx M\begin{pmatrix}\hat{k}_t\\ \hat{c}_t\end{pmatrix}. \tag{1}$$

Express the coefficients of M as a function of ρ, α, δ and η.

3 – Let $\hat{k}_0$ be the initial value of the "hatted" capital stock. Show that the only non-explosive solution to (1) is

$$\begin{aligned}\hat{k}_t &= e^{-\lambda t}\hat{k}_0,\\ \hat{c}_t &= u\hat{k}_0 e^{-\lambda t},\end{aligned}$$

where

$$\begin{aligned}\lambda &= \frac{\sqrt{\rho^2 + 4\eta(1-\alpha)(\rho+\delta)(\rho+(1-\alpha)\delta)/\alpha} - \rho}{2};\\ u &= \frac{\alpha}{\rho+(1-\alpha)\delta}\left(\rho+\lambda\right).\end{aligned}$$

[One needs to use the following mathematical result: let $V(t)$ be an $n-$vector such that $dV/dt = AV$, where A is a matrix. The general solution of this set of differential equations is a vector space of dimension n. Furthermore, if V_i is an eigenvector associated with eigenvalue λ_i, then $V(t) = e^{\lambda_i t}V_i$ is a particular solution. Consequently, we have fully characterized the solution if we have found n distinct eigenvectors.]

4 – Let $Y^\star$ be the steady state level of output and define the speed of convergence as

$$v = -\frac{\frac{d}{dt}\ln\frac{Y_t}{Y^\star}}{\ln\frac{Y_t}{Y^\star}}.$$

Show that if Y_t is close to $Y^\star$, we have

$$v \approx \lambda.$$

5 – Can you tabulate the speed of convergence for different plausible values of ρ, δ, α, and η? How does this compare with the empirical estimates of Barro and Sala-i-Martin and Mankiw Romer and Weil and with the theoretical values predicted by the Solow and Augmented Solow models?

6 – For each of these 4 parameters, say how it affects v and try to provide and economic intuition for this.

Problem II –The effect of international capital mobility on the speed of convergence

1 – Take the augmented Solow model such that

$$Y(t) = K(t)^{\alpha} H(t)^{\beta} (A(t)L)^{1-\alpha-\beta}$$

Assume L is constant and $A(t)$ grows at rate g. Compute the balanced growth path and the local speed of convergence.

2 – Assume now that this is a small open economy and that capital is internationally mobile. Which condition should replace the equation for physical capital accumulation?

3 – What is the balanced growth path of this economy?

4 – What is the local speed of convergence?

5 – How does it compare to the closed economy case?